



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PART TEST

05-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	C,D	B,C,D	B,D	A,B,D	A,D	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	C	D	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4.70 to 4.71	192.30 to 192.31	104.70 to 104.80	10.20	43.00 to 47.70	475.00	22.00	25.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,D	A,B,C	A,C,D	A,B,D	A,B,C	A,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	D	D	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	34.50 to 35.00	30.00	135.00	10.00	0.82 to 0.84	65.00	90.00	209.00 to 210.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	A,D	A,B,C	A,B,C,D	A,B,C	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	C	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.00	2.00	4.00	0.00	2.23 or 2.24	1.42 or 1.43	4.00	31.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C,D)

(A) If it convert to fully ionized form by 1.42 nm photon then

$$E = \frac{1240}{\lambda(\text{nm})} \text{ eV} = \frac{1240}{1.42} \text{ eV} = 873.23 \text{ eV}$$

$$873.28 = 13.6 \frac{z^2}{(1)^2} \Rightarrow z \simeq 8$$

(B) So atomic number not 6

$$r = 0.529 \times \frac{n^2}{z} \text{ \AA} = (0.529) \times \frac{(1)^2}{8} \text{ \AA} = 0.066 \text{ \AA}$$

(C) Angular momentum

$$= \frac{nh}{2\pi} = \frac{(1)h}{2\pi} = \frac{6.67 \times 10^{-34}}{6.28} \text{ J-s}$$

$$= 1.062 \times 10^{-34} \text{ J-s}$$

(D) If electron jump from infinite to $n = 1$. Then it will emit photon of wave length $\lambda = 1.42 \text{ nm}$.

2. Ans (B,C,D)

The total power of $P_p = \omega_0(1 - \mu)\pi R^2$ arrives from the Sun to the Earth, where R is the Earth's radius. Because the atmosphere is impenetrable for the radiation coming from the Earth then in the case of equilibrium the atmosphere has to radiate the same power outwards: $P_p = P_\alpha$ where P_α is the power radiated outwards by the atmosphere. The power radiated from the Earth is expressed as $P_m = 4\pi R^2 \sigma T_m^4$ where T_m is the ground's temperature. In the case of equilibrium it is equal to the sum of powers radiated back from the Sun and the atmosphere: $P_m = P_p + P_\alpha = 2P_p$. The Earth's temperature is expressed as :

$$T_m = \sqrt[4]{\frac{W_0 (1 - \mu)}{2\sigma}}$$

3. **Ans (B,D)**

If molar heat capacity of gas during polytropic

is $C = \frac{R}{2}$, then

$$C = C_V + \left(\frac{R}{1-x} \right)$$

$$\frac{R}{2} = \frac{5R}{2} + \frac{R}{1-x}$$

$$2R = \frac{R}{x-1}$$

$$x = 1.5$$

$$PV^{1.5} = C_1$$

$$TV^{1/2} = C_2$$

$$\text{From 3 to 4} \rightarrow (4P_0)(3V_0)^{3/2} = P_4(4V_0)^{3/2}$$

$$P_4 = \left(\frac{3\sqrt{3}}{2} \right) P_0$$

$$\text{at point 1} \rightarrow T_1 = \frac{4P_0V_0}{nR} = 300 \text{ K}$$

$$\text{at point 2} \rightarrow T_2 = \frac{4P_0V_0}{nR} = 300 \text{ K}$$

$$\text{at point 3} \rightarrow T_3 = \frac{12P_0V_0}{nR} = 900 \text{ K}$$

$$\text{at point 4} \rightarrow T_4 = \frac{6\sqrt{3}P_0V_0}{nR}$$

So, $T_{\max} = 900$ Kelvin at point 3.

4. **Ans (A,B,D)**

(A) Maximum energy of photon will be if minimum wavelength

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV}}{\lambda(\text{nm})} = \frac{1240}{365.5} \text{ eV}$$

$$E_{\max} = 3.4 \text{ eV}$$

$$E_{\min} = \frac{1240}{690.8} \text{ eV} = 1.8 \text{ eV}$$

So, these photons can emit photo-electron from cathode if work function is 1.5 eV.

(B) If the resistance of the potentiometer is adjusted to be at the left most point. Then no stopping potential so photo current will be maximum.

(D) If work function 2.39 eV. Then photon of maximum energy 3.4 eV use 2.39 eV to eject electron and remaining 1eV, K.E. will be stopped by 1 volt stopping potential.

5. **Ans (A,D)**

Work done by $PV = R(T + \alpha T^2)$ gas

In constant pressure

$$P \frac{dV}{dT} = R(1 + 2\alpha T)$$

$$PdV = R(1 + 2\alpha T)dT$$

$$W = \int_{T_1}^{T_2} P dV = \int_{T_1}^{T_2} R(1 + 2\alpha T)dT$$

$$W = R(T_2 - T_1) + \alpha R(T_2^2 - T_1^2)$$

This work done is 1.7 times by work done by ideal gas at constant pressure.

$$1.7R(T_2 - T_1) = R(T_2 - T_1)[1 + \alpha(T_1 + T_2)]$$

$$700 = T_2 + 300$$

$$T_2 = 400 \text{ Kelvin}$$

$$\text{So work done by ideal gas} = (1)R(400 - 300) = 100 R$$

$$\text{and work done by non-ideal gas} = 170 R$$

$$\Delta U = \frac{5}{2}(1)R(100) = 250 R$$

$$\Delta Q = \Delta U + 170 R = 420 R$$

6. **Ans (A,B)**

Initial mass will be same in both experiments P & Q because heat supply rate is constant and in same time in both experiments change from -ve temperature to zero is same.

Before t_0 time in experiment P, it completely melts, but in experiment Q, it completely melts after t_0 time.

In Q, it come in liquid state after taking some time, after time t_0 .

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (B)**

Due to weak nuclear force $\frac{N}{Z}$ ratio disturb, so to attain stability proton converts to neutron OR neutron converts to proton by β^+ or β^- decay.

8. Ans (C)

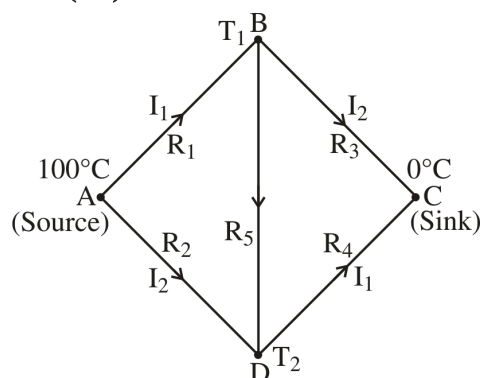
$$\text{De-Broglie wave length } \lambda = \frac{h}{p} = \frac{h}{mv}$$

So in uniform magnetic field, no work done on electron by magnetic force, so magnitude of speed and momentum will remain same.

In photo-electric tube where collector is at a positive potential then electron will accelerate, so momentum will increase and λ will decrease.

Electron in ground state of H-atom absorbs a photon and move to higher excited state in this condition. Speed and momentum decrease but λ will increase and when it return to ground state λ decrease.

9. Ans (D)



$$R_1 = 2000 ; R_2 = 10000 ; R_3 = 10000 ; R_4 = 2000 ; R_5 = 1428.57$$

$$R_5 = \frac{10000}{7} = 1428.57$$

Applying junction law at B :

$$\frac{100 - T_1}{2000} = \frac{T_1 - T_2}{1428.57} + \frac{T_1 - 0}{10000}$$

Applying junction law at D :

$$\frac{100 - T_2}{10000} + \frac{T_1 - T_2}{1428.57} = \frac{T_2 - 0}{2000}$$

On solving $T_1 = 60^\circ\text{C}$; $T_2 = 40^\circ\text{C}$

From AB :

$$\frac{(100 - 60)}{2000} = \frac{40}{2000} = \frac{4}{200} - \frac{1}{50} = 20 \text{ mW}$$

From AD :

$$\frac{100 - 40}{10000} = \frac{60}{10000} = \frac{6}{1000} = 6 \text{ mW}$$

From BD :

$$\frac{60 - 40}{\left(\frac{10000}{7}\right)} = \frac{140}{10000} = \frac{14}{1000} = 14 \text{ mW}$$

$$\text{From source} = 20 + 6 = 26 \text{ mW}$$

10. Ans (D)

Change in entropy

$$\int_{S_1}^{S_2} dS = \int \frac{dQ}{T}$$

$$\Delta S = (S_2 - S_1) = \int \frac{dQ}{T}$$

$$f_{eq} = \left(\frac{n_1 f_1 + n_2 f_2}{n_1 + n_2} \right) = \frac{(1 \times 3 + 1 \times 5)}{2} = 4$$

$$\text{For A} \rightarrow \text{B} : TV^{1/2} = C_1$$

$$\frac{PV^{3/2}}{nR} = C_1$$

$$PV^{3/2} = C_2$$

$$C = C_V + \frac{R}{1-x} = \left(\frac{4R}{2} \right) + \frac{R}{1-\frac{3}{2}} = (2R - 2R) = 0$$

$$\text{So, } dQ = nCdT = 0$$

$$\Delta S = 0 \text{ So, } S_1 = S_2$$

B to C : T = constant

and B to C it is case of compression so dw is

$$(-ve) \text{ and } dQ = dW + dU \rightarrow 0$$

dQ will be (-ve) means dS is (-ve)

$$S_2 < S_1$$

C to D : Isochoric process

$$dW = 0$$

$$dU = -ve ; dQ = -ve$$

$$\text{So, } \Delta S = -ve$$

$$S_2 < S_1$$

D to A : TV = constant

$$\frac{PV}{nR} = \text{Constant}$$

$$PV^2 = C$$

$$C = \left(\frac{4R}{2} \right) + \left(\frac{R}{1-2} \right) = 2R - R = R$$

$$dQ = nCdT < 0$$

$$\text{So, } dS < 0 ; \text{ So, } S_2 < S_1$$

Because (dT < 0)

PART-1 : PHYSICS

SECTION-II

1. Ans (4.70 to 4.71)

$$\text{Radiation pressure} = P = \frac{I}{C} \cos^2 \theta (1 + r)$$

In first case :

$$P = \frac{4}{C} (1 + 1) \cos^2 30^\circ = \frac{8}{C} \cos^2 30^\circ$$

In second case :

$$P = \frac{I}{C} (1 + 0.7) \cos^2 30^\circ = \frac{1.7 I}{C} \cos^2 30^\circ$$

$$I = \frac{8}{1.7} = 4.70 \frac{W}{m^2}$$

2. Ans (192.30 to 192.31)

The average density of the balloon is equal to the air density if the mass of the weight is as big as possible, meaning $\rho = \frac{m + M}{V}$, where m is the mass of the gas in the balloon. From this we can express $m = \rho V - M$. In addition the ideal gas law applies $pV = \frac{m}{\mu} RT$, where we get $m = \frac{pV\mu}{RT}$.

Based on these two equations we can write $\frac{pV\mu}{RT} = \rho V - M$ meaning $V = \frac{M}{\rho - \frac{p\mu}{RT}}$. For the given data $V = 193 \text{ m}^3$.

3. Ans (104.70 to 104.80)

Applying conservation of linear momentum

$$P_{P_0} = P_\alpha$$

$$P_{P_0}^2 = P_\alpha^2$$

$$2m_{P_0}k_{P_0} = 2m_\alpha k_\alpha$$

$$m_{P_0}k_{P_0} = m_\alpha k_\alpha$$

$$k_\alpha = Q + k_{P_0}$$

$$\frac{m_{P_0}k_{P_0}}{m_\alpha} = Q + k_{P_0}$$

$$k_{P_0} = Q \left(\frac{m_\alpha}{m_{P_0} - m_\alpha} \right)$$

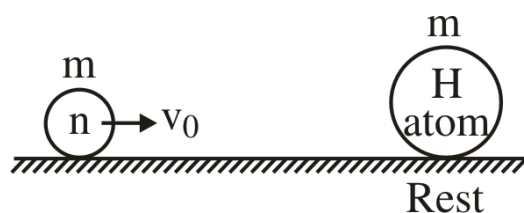
$$Q = [M_{P_0} - M_{P_b} - M_{He}] C^2$$

$$Q = 5.394 \text{ MeV}$$

$$k_{P_0} = (5.394 \times 10^3 \text{ KeV}) \left(\frac{4.00260}{205.98026} \right)$$

$$k_{P_0} = 104.80 \text{ KeV}$$

4. Ans (10.20)



Maximum loss in K.E.

$$mV_0 = (2m)V$$

$$V = \left(\frac{V_0}{2} \right)$$



$$K_i = \frac{1}{2} mV_0^2$$

$$K_f = \frac{1}{2} (2m) \left(\frac{V_0}{2} \right)^2 = \frac{mV_0^2}{4}$$

$$\Delta K = K_i - K_f =$$

$$\frac{mV_0^2}{2} - \frac{mV_0^2}{4} = \frac{mV_0^2}{4} = \frac{K_i}{2}$$

$$\text{Here, } \Delta K = \frac{24.2 \text{ eV}}{2} = 12.1 \text{ eV}$$

So, if it is partially elastic so electron by

absorbing 10.2 eV energy move from $(n = 1)$ to

$(n = 2)$.

And when come back emit photon of energy of

10.2 eV.

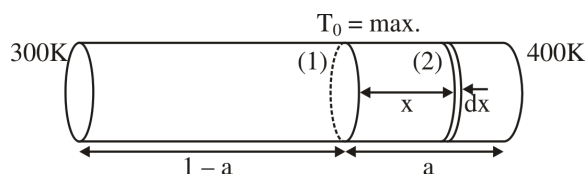
5. Ans (43.00 to 47.70)

Applying principal of calorimetry.

$$(220 \rho) S (110 - \theta) = (400 \rho) S (\theta - 10)$$

$$\theta = \frac{1410}{31} = 45.48^\circ$$

6. Ans (475.00)



$$i = \frac{UA}{\rho L}$$

$$P_{1-2} = i^2 R = \frac{U^2 A^2}{\rho^2 L^2} \times \frac{\rho x}{A} = kA \frac{dT}{dx}$$

$$\Rightarrow \frac{U^2}{\rho L^2 k} \int_0^a x dx = \int_{T_0}^{400} dT$$

$$\Rightarrow 500 a^2 = T_0 - 400$$

$$\text{Similarly, } 500 (1-a)^2 = T_0 - 300$$

$$\Rightarrow a = 0.4 \text{ m, } T_0 = 480 \text{ K}$$

For T mid point,

$$\frac{U^2}{\rho L^2 k_0} \int_0^{0.1} x dx = - \int_{480}^T dT$$

$$500 \times 0.01 = 480 - T$$

$$\Rightarrow T = 475 \text{ K}$$

7. Ans (22.00)

M _____

L _____ $E_L = 20 \text{ KeV}$

$h\nu$ $\Delta E = 42 \text{ KeV}$
K _____ $E_K = 62 \text{ KeV}$

Energy loss 42 KeV will utilized to eject electron from L using 20 KeV energy and remaining 22 KeV will be it's kinetic energy.

8. Ans (25.00)

$$N = N_0 e^{-\lambda t}$$

$$\frac{N_0}{4} = N_0 e^{-\lambda t_0}$$

$$e^{\lambda t_0} = 4$$

$$\lambda = \left(\frac{2 \log_e 2}{t_0} \right)$$

$$\frac{2N_0}{3} = N_0 e^{-\lambda t_1}$$

$$t_1 = \frac{\log_e 1.5}{\lambda}$$

$$\frac{N_0}{3} = N_0 e^{-\lambda t_2}$$

$$t_2 = \frac{\log_e 3}{\lambda}$$

$$(t_2 - t_1) = \frac{\log_e 2}{\lambda} = \frac{t_0}{2}$$

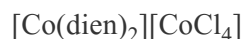
$$\text{After put } \lambda = \frac{2 \log_e 2}{t_0}$$

$$(t_2 - t_1) = 25 \text{ minute}$$

PART-2 : CHEMISTRY

SECTION-I (i)

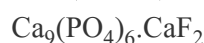
5. Ans (A,B,C)



PART-2 : CHEMISTRY

SECTION-II

4. Ans (10.00)



8. Ans (209.00 to 210.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,C,D)

$$\frac{1}{b} \begin{vmatrix} ah + bg & bg & ab + ch \\ bf + ba & bf & hb + bc \\ af + bc & bc & bg + fc \end{vmatrix} c_2 \rightarrow bc_2$$

$$C_2 \rightarrow C_2 - C_1$$

$$\frac{1}{b} \begin{vmatrix} ah + bg & -ah & ab + ch \\ bf + ba & -ab & hb + bc \\ af + bc & -af & bg + fc \end{vmatrix}$$

$$-\frac{a}{b} \begin{vmatrix} ah + bg & h & ab + ch \\ bf + ab & b & hb + bc \\ af + bc & f & bg + fc \end{vmatrix}$$

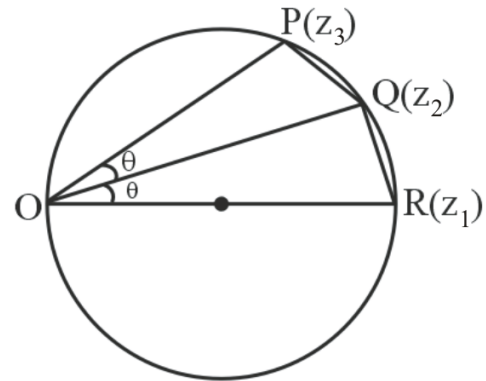
$$-\frac{a}{bc} \begin{vmatrix} ah + bg & ch & ab + ch \\ bf + ab & bc & hb + bc \\ af + bc & cf & bg + fc \end{vmatrix}$$

$$C_2 \rightarrow C_3 - C_2$$

$$-\frac{a}{bc} \begin{vmatrix} ah + bg & ch & ab \\ bf + ab & bc & hb \\ af + bc & cf & bg \end{vmatrix} = -\frac{a}{bc} \cdot bc \begin{vmatrix} ah + bg & h & a \\ bf + ab & b & h \\ af + bc & f & g \end{vmatrix}$$

2. Ans (A,D)

$$\frac{z_2}{z_1} = \frac{OQ}{OR} \cdot e^{i\theta} \dots (1)$$



$$\frac{z_3}{z_2} = \frac{OP}{OQ} e^{i\theta} \dots (2)$$

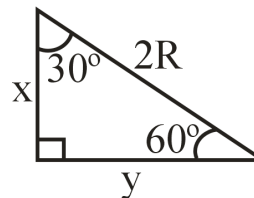
$$\frac{z_2^2}{z_1 z_3} = \frac{OQ^2}{OR \cdot OP}$$

$$\frac{z_2^2}{z_1 z_3} = \frac{\cos^2 \theta}{\cos 2\theta} = \frac{3}{2}$$

$$\theta = \frac{\pi}{6}; \angle POR = \frac{\pi}{3}$$

$$\text{Area of } \triangle OPR = \frac{1}{2} \times \sqrt{3}R \cdot R = \frac{\sqrt{3}}{2} R^2$$

$$\frac{x}{2R} = \sin 60^\circ = \frac{\sqrt{3}}{2}$$



$$\frac{y}{2R} = \cos 60^\circ = \frac{1}{2}$$

3. Ans (A,B,C)

$$AB + B^2 A^2 = I$$

$$A^3 + B^3 = 0$$

$$AB + B^2 A^2 - A^3 - B^3 = I$$

$$A(B - A^2) - B^2(B - A^2) = I; (B - A^2)(A - B^2) = I$$

$$A - B^2 \text{ is non singular, } B - A^2 \text{ is non-singular}$$

$$BA - B^3 - A^3 + A^2 B^2 = I$$

$$BA + A^2 B^2 = I$$

4. **Ans (A,B,C,D)**

$$A = \begin{bmatrix} m & n \\ p & q \end{bmatrix} \quad \text{adj} A = \begin{bmatrix} q & -n \\ -p & m \end{bmatrix}$$

$$|A| = qm - pn = d$$

$$|A + |A| \text{adj} A| = \begin{vmatrix} m + qd & n - nd \\ p - pd & q + md \end{vmatrix} = 0$$

$$(qd + m)(q + md) - (pn)(1 - d)^2 = 0$$

$$q^2d + qmd^2 + qm + m^2d - pn(1 - d)^2 = 0$$

$$d(q^2 + m^2) + (mq - pn) + (mq - pn)d^2 + 2pnd = 0$$

$$d(q^2 + m^2) + (mq - pn) + d \cdot d^2 + 2pnd = 0$$

$$d(q^2 + m^2 + 2qm) + d + d^3 - 2d^2 = 0$$

$$(q + m)^2 + (d - 1)^2 = 0 \therefore q + m = 0 \text{ and } d = 1$$

$$\therefore |A| = 1 \quad \text{Tr}(A) = 0$$

$$|A - |A| \text{adj} A| = |A - \text{adj} A|$$

$$= \begin{vmatrix} m - q & 2n \\ 2p & q - m \end{vmatrix}$$

$$-(q - m)^2 - 4pm$$

$$-[(q - m)^2 + 4(mq - 1)]$$

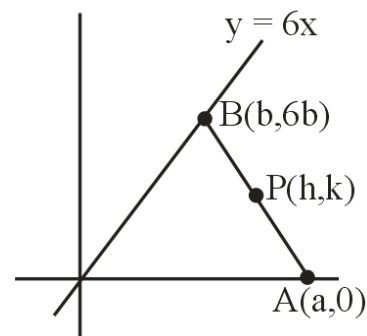
$$-[(q + m)^2 - 4] = 4$$

5. **Ans (A,B,C)**

Let $A(a,0)$

Let $B(b,6b)$

$AB = 2\ell$



$$\frac{a+b}{2} = h; \frac{6b}{2} = k$$

$$a = 2h - b \therefore b = \frac{k}{3}$$

$$a = 2h - \frac{k}{3}; b = \frac{k}{3}$$

$AB = 2\ell$

$$(a - b)^2 + 36b^2 = 4\ell^2$$

$$\left(2h - \frac{k}{3} - \frac{k}{3}\right)^2 + 36\frac{k^2}{9} = 4\ell^2$$

$$9x^2 + 10y - 6xy = 9\ell^2$$

$$\text{similarly for centroid } 9x^2 + 10y^2 - 6xy = 4\ell^2$$

$$\text{similarly } (12x - 3y)^2 + 36y^2 = 64\ell^2$$

6. **Ans (A,C)**

Any point on line L is

$$(-2 + r \cos \theta, -3 + r \sin \theta)$$

$$x + 3y = 9$$

$$-2 + r_1 \cos \theta + 3(-3 + r_1 \sin \theta) = 9$$

$$r_1 = \frac{20}{\cos \theta + 3 \sin \theta}$$

$$\text{similarly : } r_2 = \frac{4}{\cos \theta + \sin \theta}$$

$$r_1 r_2 = \frac{80}{(\cos \theta + 3 \sin \theta)(\cos \theta + \sin \theta)} = 20$$

$$4 = \cos^2 \theta + 4 \sin \theta \cos \theta + 3 \sin^2 \theta$$

$$4 \sec^2 \theta = 1 + 4 \tan \theta + 3 \tan^2 \theta$$

$$\tan^2 \theta - 4 \tan \theta + 3 = 0$$

$$\tan \theta = 1 \text{ or } 3$$

$$y + 3 = 1(x + 2) \text{ or } y + 3 = 3(x + 2)$$

$$x - y = 1 \text{ or } 3x - y = -3$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

$$(I) (1 + \omega + \omega^2 + \dots + \omega^n)^m$$

$$\left[\frac{1(\omega^{n+1} - 1)}{\omega - 1} \right]^m$$

$n = 3k$ then value 1

$n = 3k + 1$ then value $(-\omega^2)^m$

$n = 3k + 2$ then value 0

for $m = 1$ $(-\omega^2)$

for $m = 2$ $(-\omega^2)^2 = \omega$

for $m = 3$ $(-\omega^2)^3 = -1$

for $m = 4$ $(-\omega^2)^4 = \omega^2$

for $m = 5$ $(-\omega^2)^5 = -\omega$

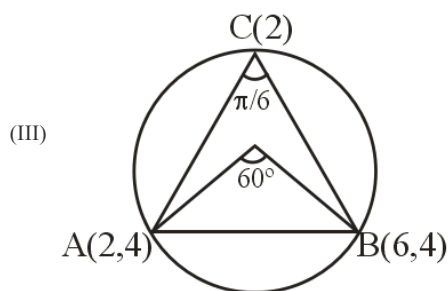
for $m = 6$ $(-\omega^2)^6 = 1$ No of distinct element = 7

(II) Root $2\omega, 2 + \omega, 2 + 3\omega^2, 2 - \omega - \omega^2$

$\therefore$ Roots are $2\omega, 2 + 3\omega, 2 + 3\omega^2, 3$

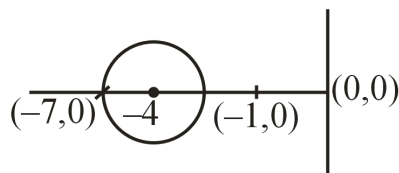
other root will be $2\bar{\omega}$

Total roots 5



Radius = 4

(IV) $|z + 4| \leq 3$ least and greatest value



maximum value is 6

8. Ans (B)

$$(I) |P| = |A|^2$$

$$\begin{vmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{vmatrix} = |A|^2$$

$$\Rightarrow 2\alpha - 6 = 4 \therefore \alpha = 5$$

$$(II) 3A^2 + 12A = \begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$$

$$\therefore B = \begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$$

$$(III) |ABA^T| = 8$$

$$|A|^2 \cdot |B| = 8 \quad \dots(1)$$

$$\det(AB^{-1}) = 8$$

$$\frac{|A|}{|B|} = 8 \quad \dots(2)$$

$$\therefore |A| = 4, |B| = \frac{1}{2}$$

$$\det(BA^{-1}B^T) = \frac{1}{2} \times \frac{1}{4} \times \frac{1}{2} = \frac{1}{16}$$

$$(IV) A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix} \quad AA^T = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix} \begin{bmatrix} 1 & \alpha \\ -\alpha & \beta \end{bmatrix}$$

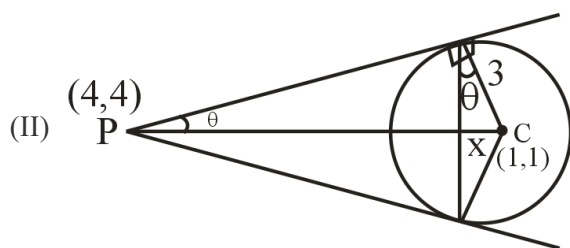
$$AA^T = \begin{bmatrix} 1 + \alpha^2 & \alpha - \alpha\beta \\ \alpha - \alpha\beta & \alpha^2 + \beta^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\alpha^2 = 0; \beta^2 = 1$$

$$\alpha^4 + \beta^4 = 1$$

9. Ans (C)

$$(I) L = \sqrt{d^2 - (r_1 + r_2)^2} = 24$$



$$PC = 3\sqrt{2}$$

$$\sin \theta = \frac{3}{3\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{x}{3} \therefore x = \frac{3}{\sqrt{2}}$$

$$\text{Length of chord} = 2\sqrt{3^2 - \frac{9}{2}} = 2\sqrt{\frac{9}{2}}$$

$$= 3\sqrt{2}$$

$$T = 2$$

$$(III) \text{ Case I : } |x - 3| = 1 \therefore x = 2 \text{ or } 4$$

$$|x - 3| \neq 0 \text{ and } 3x^2 - 10x + 3 = 0$$

$$(3x - 1)(x - 3) = 0$$

$$x = \frac{1}{3}; x = 3 \text{ (reject)}$$

$$\therefore x = 2, 4, \frac{1}{3}$$

$$(IV) \sqrt{2 + \sqrt{5} - (\sqrt{5} - 2)} = 2$$

10. Ans (A)

$$(I) k > |7i - 2i| \text{ or } k > 5$$

$$(II) x^2 - x + 1 = 0 \therefore x = -\omega, -\omega^2$$

$$\left(x^n + \frac{1}{x^n}\right)^2 = \left((- \omega)^n + \frac{1}{(- \omega)^n}\right)^2$$

$$= (-1)^{2n} \cdot (\omega^{2n} + \omega^n)^2$$

$$1 + \omega^n + \omega^{2n} = 0; n \neq 3p$$

$$= 2; n = 3p$$

$$4(-1)^2 + (2)^2 = 8$$

$$(III) \text{ Let } (z - 1) = re^{i\theta}$$

$$\therefore e^{i\theta} \cdot e^{-i\alpha} + \frac{1}{re^{i(\theta - \alpha)}}$$

$$\therefore re^{i(\theta - \alpha)} + \frac{1}{re^{i(\theta - \alpha)}}$$

$$z_0 + \frac{1}{z_0} \text{ is purely real}$$

$$\therefore |z_0| = 1 \therefore |z - 1| = 1$$

$$(IV) (xy - 7)^2 = x^2 + y^2$$

$$x^2y^2 - 14xy + 49 = x^2 + y^2$$

$$x^2y^2 - 12xy + 49 = (x + y)^2$$

$$(xy - 6)^2 + 13 = (x + y)^2$$

$$(x + y)^2 - (xy - 6)^2 = 13$$

Now we know difference of square of two consecutive integers

$$7^2 - 6^2 = 13$$

$$\therefore x + y = 7; xy - 6 = \pm 6$$

$$xy = 12, 0$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (0.00)**

$$T_p, T_{2q}, T_{3r}$$

a, b, c in H.P.

$$\frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ in A.P.}$$

$$T_p = A + (p-1)d = \frac{1}{a}$$

$$T_{2q} = A + (2q-1)d = \frac{1}{b}$$

$$T_{3r} = A + (3r-1)d = \frac{1}{c}$$

$$\text{Now } abc \begin{vmatrix} \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ p & 2q & 3r \\ 1 & 1 & 1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 - (A-d)R_3 - dR_2$$

$$abc \begin{vmatrix} 0 & 0 & 0 \\ p & 2q & 3r \\ 1 & 1 & 1 \end{vmatrix} = 0$$

2. **Ans (2.00)**

$$\Delta = 2(\tan A + \tan B + \tan C)^3 \cdot \tan A \tan B \tan C$$

$$\Delta = 2(\tan A + \tan B + \tan C)^4$$

$$\text{minimum value of } \tan A + \tan B + \tan C = 3\sqrt{3}$$

$$\therefore \frac{\Delta}{1000} \geq 1.458$$

$$\text{least value} = 2$$

3. **Ans (4.00)**

$$C_1 : \text{centre } (1, 8) ; \text{radius} = 1$$

$$C_2 : \text{centre } (1, -1) ; \text{radius} = 2$$

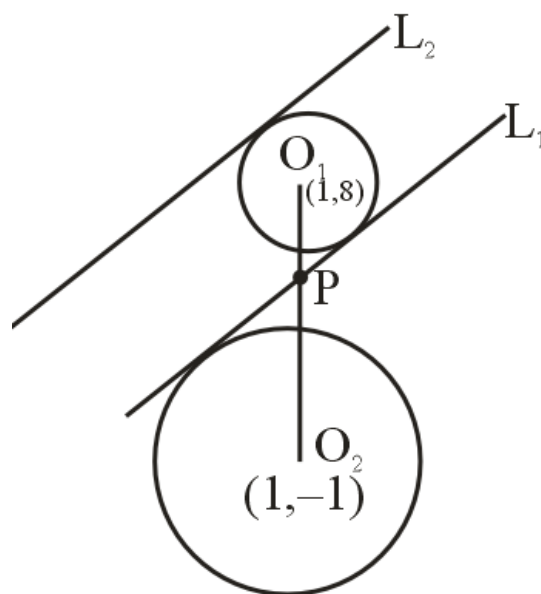
$$O_1P : O_2P = \frac{1}{2}$$

$$\therefore P(1, 5)$$

Equation of L_1 is

$$y - 5 = m(x - 1)$$

$$L_1 : mx - y - m + 5 = 0$$



L_1 is tangent of circle C_2

$$\left| \frac{m(1) - (-1) - m + 5}{\sqrt{m^2 + 1}} \right| = 2$$

$$6 = 2\sqrt{m^2 + 1} \therefore m^2 = 8 \therefore m = 2\sqrt{2}$$

$$\therefore 2\sqrt{2}x - y - 2\sqrt{2} + 5 = 0$$

similarly equation of L_2 is $2\sqrt{2}x - y + k = 0$

$$\left| \frac{2\sqrt{2} - 8 + k}{3} \right| = 1$$

$$\therefore k = \pm 3 + 8 - 2\sqrt{2}$$

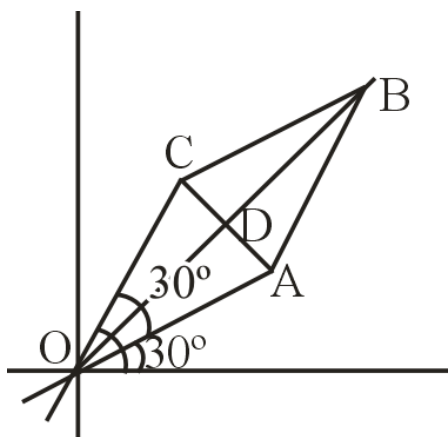
$$\therefore k = 11 - 2\sqrt{2}$$

4. Ans (0.00)

$$\angle COA = 30^\circ$$

$$OA = OC = 2, \angle OAB = 150^\circ$$

$$\cos 150^\circ = \frac{OA^2 + AB^2 - OB^2}{2OA \cdot AB}$$



$$\frac{-\sqrt{3}}{2} = \frac{4 + 4 - OB^2}{2 \cdot 2 \cdot 2}$$

$$-4\sqrt{3} = 8 - OB^2$$

$$OB^2 = 8 + 4\sqrt{3}$$

$$OB = \sqrt{2}(\sqrt{3} + 1)$$

coordinates of B

$$\Rightarrow (\pm\sqrt{2}(\sqrt{3} + 1) \cos 45^\circ, \pm\sqrt{2}(\sqrt{3} + 1) \sin 45^\circ)$$

5. Ans (2.23 or 2.24)

$$4a^2 - 5b^2 + 6a + 1 = 0$$

$$9a^2 + 6a + 1 = 5(a^2 + b^2)$$

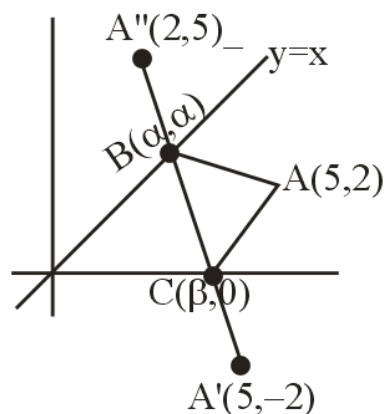
$$(3a + 1)^2 = 5(a^2 + b^2)$$

$$\left| \frac{3a + 1}{\sqrt{a^2 + b^2}} \right| = \sqrt{5}$$

$$\left| \frac{a(3) + b(0) + 1}{\sqrt{a^2 + b^2}} \right| = \sqrt{5}$$

Radius is $\sqrt{5}$

6. Ans (1.42 or 1.43)



$$AB = A''B \text{ and } AC = A'C$$

$$AB + BC + AC = \text{minimum}$$

$$\text{or } A''B + BC + A'C = \text{Min}$$

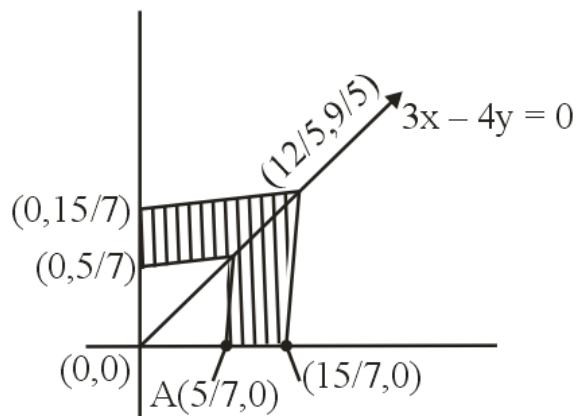
$$\text{Line } AA'' = y - 5 = -\frac{7}{3}(x - 2) \therefore 3y - 15 = -7x + 14$$

$$7x + 3y = 29$$

$$\alpha = \frac{29}{10}, \beta = \frac{29}{7}$$

7. Ans (4.00)

$$1 \leq \frac{|3x - 4y|}{5} + \frac{|4x + 3y|}{5} \leq 3$$



$$5 \leq |3x - 4y| + |4x + 3y| \leq 15$$

$$5 \leq |3x - 4y| + 4x + 3y \leq 15$$

$$\text{Case I : } 3x - 4y \geq 0$$

$$5 \leq 3x - 4y + 4x + 3y \leq 15$$

$$5 \leq 7x - y \leq 15$$

$$\text{Case II : } 3x - 4y \leq 0$$

$$5 \leq -3x + 4y + 4x + 3y \leq 15$$

$$5 \leq x + 7y \leq 15$$

$$A_1 = \text{Area}$$

$$= \frac{1}{2} \times \frac{15}{7} \times \frac{9}{5} - \frac{1}{2} \times \frac{5}{7} \times \frac{3}{5} = \frac{12}{7}$$

$$A_2 = \text{Area} =$$

$$\frac{1}{2} \times \frac{15}{7} \times \frac{12}{5} - \frac{1}{2} \times \frac{5}{7} \times \frac{4}{5} = \frac{36}{14} - \frac{4}{14} = \frac{16}{7}$$

$$\therefore \text{Required Area} = \frac{12}{7} + \frac{16}{7} = 4$$

8. Ans (31.00)

$$\left| \frac{z - 4i}{z - 2i} \right| = 1$$

$\Rightarrow |z - 4i| = |z - 2i|$ thus z represents a point equidistant from the points $2i$ and $4i$

$\therefore z$ lies on perpendicular bisector of line

joining $2i$ and $4i \therefore z = x + 3i$

$$\left| \frac{z - 8 + 3i}{z + 3i} \right| = \frac{3}{5}$$

$$5|x - 8 + 6i| = 3|x + 6i|$$

$$\text{solving } x^2 - 25x + 136 = 0$$

$$(x - 8)(x - 17) = 0$$

$$x = 8, 17$$

$$\therefore z = 8 + 3i; z = 17 + 3i$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PART TEST

05-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	D	D	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,D	A,B,D	A	D	A,B,D	A,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	5	4	5	5	2	4	4

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	B	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,C,D	A,B,D	A,C,D	A,B,C,D	A,B,D	A,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	2	3	6	2	2	2	5	8

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	D	A	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C,D	A,B,C	A,B,D	A,C,D	A,C,D	A,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	1	8	6	5	4	8	3	5

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)

$$X^{11} \rightarrow X^{10} + n'$$

$$BE_i = BE_f + \Delta E$$

$$(7.8) \times (11) = (8)(10) + \Delta E$$

2. Ans (D)

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1 + R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{\text{total}}} = \frac{1}{\frac{L}{k_1 A} + \frac{L}{k_2 A} + \frac{2L}{k_3 \frac{A}{n}}}$$

$$= \frac{k_1 k_2 A}{L(k_1 + k_2)} + \frac{k_3 A}{2Ln}$$

$$= \frac{2k_1 k_2 n + k_3(k_1 + k_2)}{2n(k_1 + k_2)} \frac{A}{L}$$

3. Ans (D)

$$A = A_0 e^{-\lambda t}$$

$$\frac{A}{A_0} = e^{-\frac{\ell_{n2}}{T} t}$$

$$\frac{A}{A_0} = e^{-(\ell_{n2}) \frac{t}{T}}$$

4. Ans (B)

Surface energy $\propto$ Area& rate of heat flow $\propto \frac{1}{\text{Area}}$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,D)

$$Q_1 = mS (T_1 - T)$$

$$Q_3 = mL$$

$$Q_3 = \frac{mS}{2}(T - T_2)$$

all heats are equal (given)

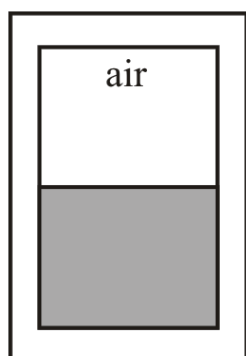
$$(T_1 - T) = \frac{1}{2}(T - T_2)$$

$$2T_1 - 2T = T - T_2$$

$$T = \left(\frac{2T_1 + T_2}{3} \right)$$

$$S (T_1 - T) = L$$

$$S \left(\frac{T_1 - T_2}{3} \right) = L$$



$$v_1 = v_0 - kv_0$$

$$= v_0 (1 - k)$$

$$v_2 = v_0 - \frac{kv_0}{\lambda}$$

$$= v_0 \left(1 - \frac{k}{\lambda} \right)$$

$$v' = \frac{m}{\lambda \rho_0}$$

$$= \frac{kv_0}{\lambda}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right) \left(\frac{T_2}{T_1} \right)$$

$$\frac{(1 - k) \lambda}{\lambda - k} \left(\frac{T_2}{T_1} \right)$$

6. Ans (A,B,D)

$$E_K = \frac{hC}{\lambda_1}$$

$$E_L = \frac{hC}{\lambda_2}$$

$$E_{K\alpha} = E_K - E_L$$

$$\frac{hC}{\lambda_{K\alpha}} = hC \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)$$

$$\frac{1}{\lambda_{K\alpha}} = \frac{1}{0.25 \text{ nm}} - \frac{1}{1 \text{ nm}}$$

$$\Rightarrow \lambda_{K\alpha} = \frac{1}{3} \text{ nm}$$

$$\frac{1}{\lambda_{K\alpha}} = R(Z - 1)^2 \left(1 - \frac{1}{4} \right)$$

$$3 \times 10^9 = 10^7 (Z - 1)^2 \left(\frac{3}{4} \right)$$

$$\Rightarrow Z = 21$$

7. Ans (A)

$$BE_H = (13.6 \text{ eV}) \frac{1^2}{n^2} \text{ for H atom}$$

$$BE_{Li} = 200. (13.6 \text{ eV}) \frac{3^2}{n^2} \text{ for Li mesic ion}$$

$$r \propto \frac{n^2}{Zm} = \frac{r_B}{600}$$

$$v \propto \frac{Z}{n}$$

$$mvr = \frac{nh}{2\pi}$$

8. Ans (D)

$$V_0 = \left(\frac{hC}{\lambda} - \phi \right) \frac{1}{e}$$

$$\Rightarrow V_{01} > V_{02} > V_{03} \text{ \& } V_0 \propto \frac{1}{\lambda}$$

$$V_0 = (hv - \phi) \frac{1}{e} \Rightarrow V_0 \propto v \text{ and}$$

slope of straight line is $\frac{h}{e}$ = same for all metals.

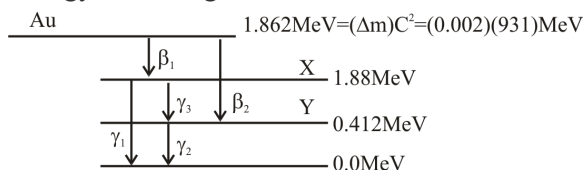
$$\phi = \frac{hC}{e\lambda_0} \Rightarrow \phi \propto \frac{1}{\lambda_0}$$

9. Ans (A,B,D)

Refer class notes of β^+ decay and K capture.

10. Ans (A,C)

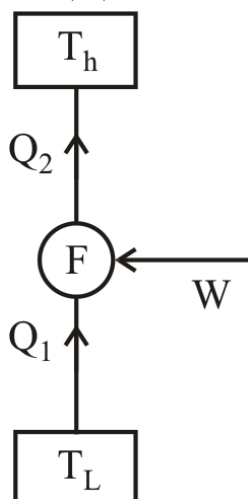
Energy level diagram



PART-1 : PHYSICS

SECTION-III

1. Ans (3)



$$Q_2 = Q_1 + W$$

$$\frac{Q_2}{Q_1} = 1 + \frac{W}{Q_1}$$

$$\frac{T_h}{T_L} = 1 + \frac{W}{Q_1}$$

$$\frac{299}{260} = 1 + \frac{W}{20 \text{ kJ}}$$

$$W = 3 \text{ kJ/s}$$

2. Ans (5)

$$E = \frac{9 \times 10^9 \times 4 \times 1.6 \times 10^{-19}}{(1.2 \times 10^{-15} \times 2)^2}$$

$$= \frac{9 \times 6.4 \times 10^{-10}}{2.4 \times 2.4 \times 10^{-30}} = 10^{21}$$

3. Ans (4)

$$m_{\text{atom}} = 1.6 \times 10^{-27} \text{ kg}$$

$$\frac{hc}{\lambda} = mvc$$

$$hcR \left(\frac{1}{n^2} - \frac{1}{m^2} \right) = mvc$$

$$6.4 \times 10^{-34} \times 10^7 \left(\frac{1}{n^2} - \frac{1}{m^2} \right) = 1.6 \times 10^{-27} \times \frac{7}{36}$$

$$\frac{1}{n^2} - \frac{1}{m^2} = \frac{7}{144} = \frac{1}{9} - \frac{1}{16}$$

4. Ans (5)

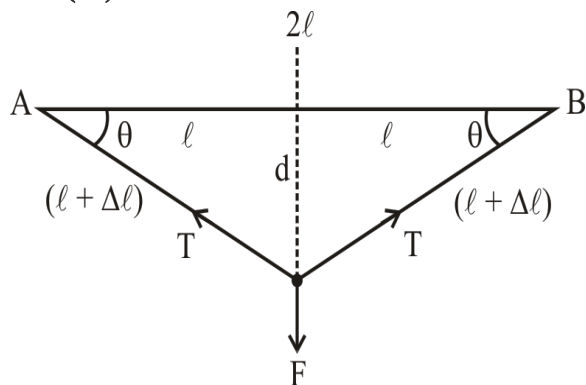
Because the gas tank is cylindrical then the pressures in the direction of its axis and the ones perpendicular to it are different. In both cases the force in the tank's wall has to balance the force caused by the pressure of the gas inside the tank. For the pressure with the direction of the axis we get an equation $\pi r^2 p_1 = 2\pi r t \sigma$ meaning $p_1 = \frac{2t\sigma}{r}$. We find the pressure with the perpendicular direction to the tank's axis by observing the imaginary rectangle-shaped cut that splits the cylinder. From the force balance we get $2rhp_2 = 2ht\sigma$ where $h \gg r$ is the tank's height, meaning $p_2 = \frac{t\sigma}{r}$. The pressure endured by the tank is smaller from these two, meaning

5. Ans (5)

The stove is cast with $n = m/\mu$ moles of water that during vaporizing would create an additional pressure in the hermetically closed room, it can be found from the ideal gas law: $\Delta p = nRT/V$. This pressure applies a force $F = \Delta p A$ to the door. This force is applied evenly all over the surface of the door, therefore when calculating the torque applied to the door we have to take half of the door's width as the force arm. The sauna visitors applied their force F_1 to the handle when holding the door, the force arm in this case is the total width of the door. Because of this we can say that the sauna visitors would have to apply the following force to hold the door:

$$F_1 = \frac{F}{2} = \frac{mRTA}{2\mu V} = 5 \text{ kN}$$

6. Ans (2)



at equilibrium $F = 2 T \sin \theta$

$$T = \frac{Ay (\Delta \ell)}{\ell}$$

$$(\ell + \Delta \ell)^2 = d^2 + \ell^2$$

$$\ell^2 \left(1 + \frac{\Delta \ell}{\ell}\right)^2 = d^2 + \ell^2$$

$$\left(1 + \frac{\Delta \ell}{\ell}\right)^2 = 1 + \frac{d^2}{\ell^2}$$

Binomial approx.

$$1 + \frac{2\Delta \ell}{\ell} = 1 + \frac{d^2}{\ell^2}$$

$$\Rightarrow \Delta \ell = \frac{d^2}{2\ell}$$

$$= \frac{(1\text{cm})^2}{2(1\text{m})}$$

$$= 0.5 \times 10^{-4} \text{ m}$$

$$\sin \theta = \frac{d}{\ell + \Delta \ell} = 0.01$$

$$T = \frac{Ay}{\ell} \Delta \ell$$

$$= \frac{(1\text{mm})^2 (2 \times 10^{11} \text{ N/m}^2)}{(1\text{m})} (0.5 \times 10^{-4} \text{ m})$$

$$= 10 \text{ N}$$

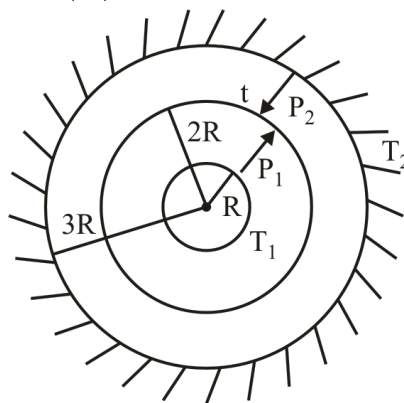
$$F = 2T \sin \theta$$

$$= 2(10)(0.01)$$

$$= 0.2 \text{ N}$$

$$\Rightarrow 10F = 2\text{N}$$

7. Ans (4)



$$P_1 = \sigma (4\pi R^2) (T_1^4)$$

$$P_2 = \sigma (4\pi (4R^2)) (T_2^4)$$

$$P_{\text{net}} = \sigma 4\pi R^2 (t^4) + \sigma 4\pi (4R^2) t^4$$

$$T_1^4 + 4T_2^4 = 5t^4$$

$$t = \left(\frac{T_1^4}{5} + \frac{T_2^4}{5} \right)^{\frac{1}{4}}$$

$$x = 5, y = \frac{5}{4} \Rightarrow \frac{x}{y} = 4$$

8. Ans (4)

Heat lost by cylinder = Heat gained by ice

$$\Rightarrow MS(50) = M_{\text{ice}}(80)$$

∴ For cylinders A and B, if ice melted is

$$m_{\text{iceA}} \text{ and } m_{\text{iceB}}, \text{ then, } \frac{S_A}{S_B} = \frac{m_{\text{iceA}}}{m_{\text{iceB}}}$$

$$\Rightarrow 2 \times h_B a_B \rho_{\text{ice}} = h_A a_A \rho_{\text{ice}} \dots (1)$$

∴ Heights of both cylinders were same say 'H'.

$$\therefore m = a_A H \rho_A = a_B H \rho_B \Rightarrow \frac{a_B}{a_A} = 2$$

$$\therefore \text{From (1), } \frac{h_A}{h_B} = 2 \times 2 = 4$$

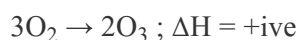
PART-2 : CHEMISTRY

SECTION-I (ii)

10. Ans (A,C,D)

Y – O₃ (Ozone)

X – O₂ (Dioxygen)



PART-2 : CHEMISTRY

SECTION-III

1. **Ans (2)**

$x = \text{Fe}, \text{Zn}, \text{Co}, \text{Ca}, \text{Pb}$

$y = \text{Ag}, \text{Au}, \text{Pt}$

6. **Ans (2)**

$X = [\text{Ni}(\text{H}_2\text{O})_4\text{en}]^{2+}$

$Y = [\text{Ni}(\text{H}_2\text{O})_2(\text{en})_2]^{2+}$

and

$Z = [\text{Ni}(\text{en})_3]^{2+}$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

$$x^7 - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2)(x - \alpha_3)(x - \alpha_4)(x - \alpha_5)$$

$$(x - \alpha_6)$$

$$\frac{x^7 - 1}{x - 1} = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3)(x - \alpha_4)(x - \alpha_5)(x - \alpha_6)$$

...(i)

$$\text{at } x = \omega : \frac{\omega^7 - 1}{\omega - 1}$$

$$= (\omega - \alpha_1)(\omega - \alpha_2)(\omega - \alpha_3)(\omega - \alpha_4)(\omega - \alpha_5)(\omega - \alpha_6)$$

...(ii)

similarly at $x = \omega^2$ in (1), we get

$$\frac{\omega^{14} - 1}{\omega^2 - 1} = (\omega^2 - \alpha_1)(\omega^2 - \alpha_2)(\omega^2 - \alpha_3)(\omega^2 - \alpha_4)(\omega^2 - \alpha_5)(\omega^2 - \alpha_6)$$

...(iii)

$$\frac{\text{ii}}{\text{iii}} \Rightarrow \prod_{i=1}^6 \frac{\omega - \alpha_i}{\omega^2 - \alpha_i} = 1 \quad (\because \omega^3 = 1)$$

2. **Ans (D)**

Let $z = x + iy$

$$-8((x + iy)^2 + (x - iy)^2) + 20(x^2 + y^2) = 288$$

$$-8(2(x^2 - y^2)) + 20(x^2 + y^2) = 288$$

$$4x^2 + 36y^2 = 288$$

$$\frac{x^2}{72} + \frac{y^2}{8} = 1$$

$$a = 6\sqrt{2}, b = 2\sqrt{2}$$

$$\text{Area} = \pi ab = 24\pi$$

3. **Ans (A)**

$$a^2 - b^2 = c^2 + 2$$

where c is a even natural number

$$\Rightarrow (a + b)(a - b) = c^2 + 2$$

(i) if $a, b \in \text{odd} \Rightarrow (a + b)(a - b)$ is divisible by 4, but $c^2 + 2$ is not divisible by 4 so case(i) rejected

(ii) if $a, b \in \text{even} \Rightarrow$ case is same as (i)

so case (ii) is rejected

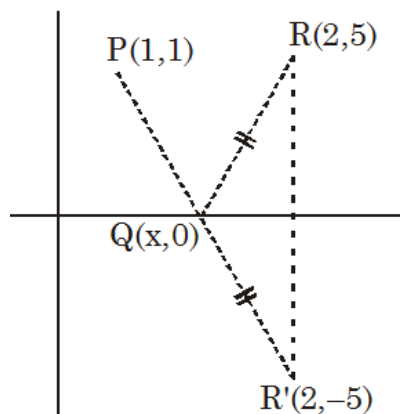
(ii) if a, b are even and odd respectively

$$\Rightarrow (a + b)(a - b) \text{ is an odd number}$$

but $c^2 + 2$ is even

so case (iii) rejected.

4. **Ans (B)**



$$f(x) = \sqrt{(x-1)^2 + (0-1)^2} + \sqrt{(x-2)^2 + (0-5)^2}$$

$$f(x) = PQ + QR$$

$\therefore f(x)$ will be minimum if P, Q, R' are collinear.

$$\therefore f(x) = PR' = \sqrt{1+36} = \sqrt{37} = m$$

$$\& \text{ equation of line } PR' \text{ is } 6x + y = 7 \Rightarrow \alpha = \frac{7}{6}$$

$$\Rightarrow [m] = 6 \& [\alpha] = 1$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,B,C,D)**

Given, $|2z + 3i| = |z|^2$

$$\therefore |2z + 3i| \leq 2|z| + 3$$

$$\Rightarrow |z|^2 \leq 2|z| + 3$$

$$\Rightarrow 0 \leq |z| \leq 3 \quad \dots(1)$$

Again, $|2z + 3i| \geq |2|z| - 3|$

$$\Rightarrow |z|^2 \geq |2|z| - 3|$$

$$\therefore |z| \geq 1 \quad \dots(2)$$

So, $(1) \cap (2)$ gives

$$1 \leq |z| \leq 3$$

Also, $|z|_{\text{maximum}} \Rightarrow z = 3i$

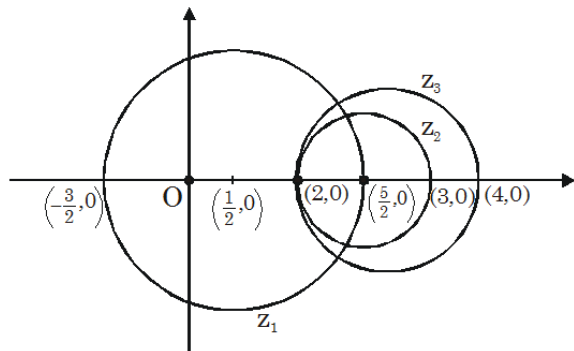
So, $a = 0, b = 3$

& $|z|_{\text{minimum}} \Rightarrow z = -i$

so, $x = 0, y = -1$

6. **Ans (A,B,C)**

$$\left| z_1 - \frac{1}{2} \right| = 2; \left| z_2 - \frac{5}{2} \right| = \frac{1}{2}; |z_3 - 3| = 1$$



from figure,

$$|z_1 + z_2 + z_3|_{\text{max}} = 4 + 3 + \frac{5}{2} = \frac{19}{2}$$

$$|z_1 + z_2 + z_3|_{\text{min}} = 2 + 2 - \frac{3}{2} = \frac{5}{2}$$

$$|z_1 - z_2|_{\text{max}} = \frac{9}{2}$$

$$|z_2 - z_3|_{\text{min}} = 0$$

7. **Ans (A,B,D)**

$$\sqrt{12 + 2\sqrt{15} + 4\sqrt{3} + 4\sqrt{5}} = 2 + \sqrt{5} + \sqrt{3}$$

$$\therefore a = 16, b = 25, c = 9$$

$$a + b + c = 50$$

8. **Ans (A,C,D)**

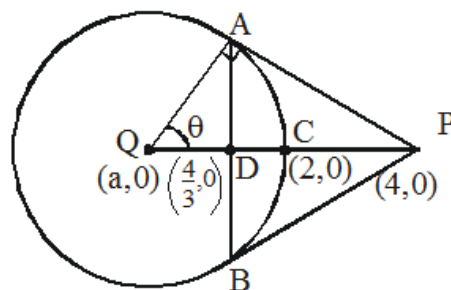
$$D = \begin{vmatrix} 2 & p^2 & 6 \\ 1 & 2 & 2q \\ 1 & 1 & 3 \end{vmatrix} = (p^2 - 2)(2q - 3)$$

$$D_2 = 0$$

$$D_1 = (p^2 - 2)(8q - 15), D_3 = (p^2 - 2)$$

9. **Ans (A,C,D)**

In ΔAQD , $AQ = AC = 2 - a$



$$\cos \theta = \frac{QD}{AD} = \frac{\frac{4}{3} - a}{2 - a} \quad \dots(i)$$

in ΔPAQ

$$\cos \theta = \frac{AQ}{QP} = \frac{2 - a}{4 - a} \quad \dots(ii)$$

from (i) and (ii)

$$\frac{\frac{4}{3} - a}{2 - a} = \frac{2 - a}{4 - a} \Rightarrow a = 1 \Rightarrow Q(1, 0)$$

$\Rightarrow$ radius $r = 1$

$$\frac{\text{area of } \Delta PAB}{\text{area of } \Delta AQB} = \frac{PQ}{QD} = \frac{4 - \frac{4}{3}}{\frac{4}{3} - 1} = 8$$

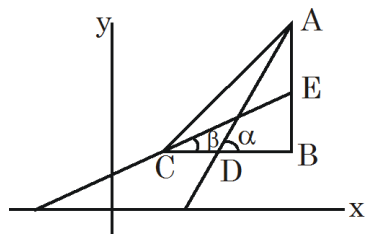
$$PA = \sqrt{PQ^2 - AQ^2}$$

$$= \sqrt{9 - 1} = 2\sqrt{2}$$

10. Ans (A,C,D)

Let slope of median AD and CE are respectively m_1 and m_2 {where $m_1 = 1$ }

$$\tan \alpha = \frac{c}{\frac{a}{2}} = \frac{2c}{a}$$



$$\tan \beta = \frac{c}{2a} = \frac{c}{2a}$$

$$\frac{m_1}{m_2} = \frac{\tan \alpha}{\tan \beta} = 4 \therefore m_2 = \frac{1}{4}$$

Let θ be angle

$$\tan \theta = \left| \frac{1 - \frac{1}{4}}{1 + \frac{1}{4}} \right| = \frac{3}{5}$$

$$\theta = \tan^{-1} \left(\frac{3}{5} \right)$$

If $BC = 2 \Rightarrow BD = CD = 1$ & $AB = 1$

$$\Rightarrow AD = \sqrt{2}$$

$$\therefore GD = \frac{\sqrt{2}}{3}$$

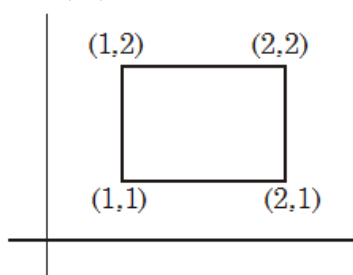
Let D is $(\alpha, \alpha - 1)$

$$\therefore GD = \frac{\sqrt{2}}{3} \text{ gives } \alpha = 2 - \frac{\sqrt{2}}{3}$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (1)



length of diagonal $= \sqrt{2} = 2r$

$$\therefore r = \frac{1}{\sqrt{2}}$$

$$\text{Area of circle } A = \pi r^2 = \frac{\pi}{2}$$

$$\therefore [A] = 1$$

2. Ans (8)

For consistency,

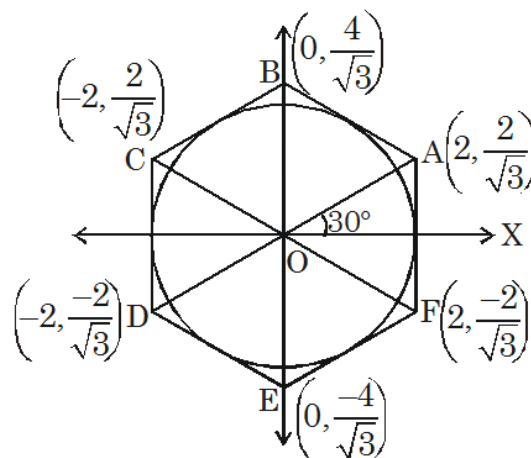
$$\begin{vmatrix} a & 1 & b \\ b & 1 & a \\ a & b & ab \end{vmatrix} = 0$$

$$\Rightarrow (a - b)(a - b^2) = 0$$

$\therefore$ Possible ordered pairs are

$$(0, 0), (1, 1), (2, 2), (3, 3), (4, 4), (4, 2), \{9, 9\}, \{9, 3\}$$

3. Ans (6)



Shifting origin to $(1, \frac{1}{\sqrt{3}})$ and $A(3, \sqrt{3})$

becomes $A(2, \frac{2}{\sqrt{3}})$. F is image of A about

x-axis. C is image of F about origin. D is image of A about origin

$\Rightarrow$ actual vertices will be $A(3, \sqrt{3})$,

$B(1, \frac{5}{\sqrt{3}})$, $C(-1, \sqrt{3})$, $D(-1, \frac{-1}{\sqrt{3}})$,

$E(1, -\sqrt{3})$, $F(3, -\frac{1}{\sqrt{3}})$

$$\Rightarrow \sum (x_i - y_i) = 3 - \sqrt{3} \Rightarrow a = 3, b = 3$$

4. Ans (5)

$$AX = \lambda X \Rightarrow (A - \lambda I)X = 0$$

$$\text{but } X \neq 0 \Rightarrow |A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 4 - \lambda & 6 & 6 \\ 1 & 3 - \lambda & 2 \\ -1 & -5 & -2 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 2)^2 = 0 \Rightarrow \lambda = 1, 2, 2$$

Sum of distinct values $1 + 2 + 2 = 5$

5. Ans (4)

The largest integer in the given interval is

$$\left[\frac{3x}{10} \right]$$

If we put $\frac{3x}{10} = y$ then $[y] - 3 < \frac{y}{3} \leq [y] - 2$ solving;

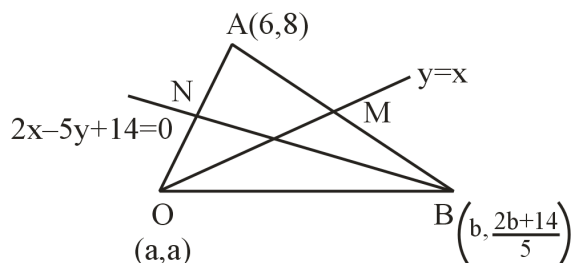
$$y \in \{3\} \cup [4, 5) \Rightarrow x \in \{10\} \cup \left[\frac{40}{3}, \frac{50}{3} \right)$$

So the possible integral values of x are 10, 14, 15, 16

$$\therefore I = 16$$

$$\therefore \frac{I}{4} = 4$$

6. Ans (8)



$\therefore$ coordinate of $N \left(\frac{a+6}{2}, \frac{b+8}{2} \right)$ satisfy

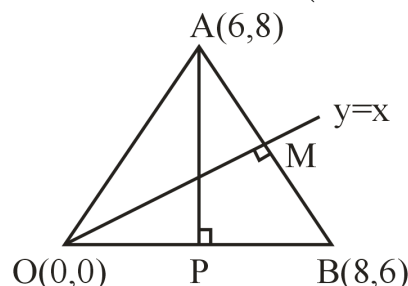
$$2x - 5y + 14 = 0$$

& coordinate of $M \left(\frac{b+6}{2}, \frac{\frac{2b+14}{5} + 8}{2} \right)$

satisfy $y = x$

This gives $O(0, 0)$ & $B(8, 6)$

$\therefore$ Centroid of $\triangle ABC$ is $\left(\frac{14}{3}, \frac{14}{3} \right)$



Altitude through vertex A is $4x + 3y = 48$

& altitude through vertex O is $y = x$

$\therefore$ Orthocentre of $\triangle ABC$ $\left(\frac{48}{7}, \frac{48}{7} \right)$

$$\therefore \frac{7(\alpha + \beta)}{12} = 8$$

7. Ans (3)

$$|1 - x|^{\log_2(x+1)} > |x - 1|^{\log_2(4-x)}; x \in (-1, 4)$$

$$\text{Case I : } |x - 1| > 1 \Rightarrow x \in (-\infty, 0) \cup (2, \infty)$$

$$\log_2(x+1) > \log_2(4-x) \Rightarrow x > \frac{3}{2}$$

$$\therefore x \in (2, 4)$$

$$\text{Case II : } |x - 1| < 1 \Rightarrow x \in (0, 2)$$

$$\therefore \log_2(4-x) > \log_2(1+x)$$

$$\Rightarrow x < \frac{3}{2}$$

$$\therefore x \in \left(0, \frac{3}{2} \right)$$

but $x = 1$ does not satisfy

$$\therefore \text{final Ans. : } x \in \left(0, \frac{3}{2} \right) \cup (2, 4) - \{1\}$$

$\therefore$ only integer value = 3

8. Ans (5)

$$\text{adj}A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix}, |A| > 0$$

$$|\text{adj}A| = |A|^2 = 16$$

$$\text{or } |A| = 4$$

$$\therefore \text{adj}(\text{adj}A) = |A|.A = 4A$$

then diagonal elements of A

$$A = \frac{1}{4} \begin{bmatrix} 8 & & \\ & 8 & \\ & & 4 \end{bmatrix} \Rightarrow \text{tr}(A) = 5$$